SHIRUI WEI

National Astronomical Observatories, Chinese Academy of Sciences (NAOC)

weisr@bao.ac.cn | ruinov1st.github.io | github.com/ruinov1st | (+86) 18007712049

Research Interests: AI-driven analysis of large-scale astronomical survey data, with a focus on multimodal deep learning methods for measuring galaxy physical properties.

EDUCATION

University of Chinese Academy of Sciences (UCAS) — Sep. 2023 - Jun. 2028
National Astronomical Observatories (NAOC)
PhD student in Astronomical Techniques and Methodology — GPA: 3.97/4.0
Supervisors: Prof. Chenzhou Cui & Dr. Changhua Li

Max Plank Institute for Astronomy (MPIA) — Jun. 2025 - Sep. 2025
Summer student | Host Supervisor: Dr. Ivelina Momcheva

Nankai University — Aug. 2019 - Jun. 2023
B.E in Software Engineering — GPA: 3.81/4.0

CURRENT RESEARCH

Automated Measurement of Physical Parameters of Large Galaxy Samples and Scientific Applications — NAOC | Aug. 2024 – Present
`PhD Thesis` | Supervisors: Prof. Chenzhou Cui, Dr. Changhua Li, Prof. Yanxia Zhang
Develop multimodal deep learning models to improve measurements of galaxy physical properties, and apply these methods to data from large-scale surveys (DESI, CSST, and Euclid) to produce value-added catalogs (VACs) and analysis pipelines for downstream scientific studies, enabling investigations of galaxy evolution, including scaling relations and environmental and redshift evolution.

Dwarf Galaxies in the JWST/OutThere Survey — MPIA | Jun. 2025 – Present
`Full-time Summer Internship` | Supervisors: Dr. Ivelina Momcheva, Dr. Raphael Hviding
Investigate the physical properties and environments of four dwarf galaxies identified in the OutThere survey, representing the first application of the surface brightness fluctuation (SBF) technique to JWST observations.

PUBLICATIONS

First Author Papers:

- **Shirui Wei** et al. (2026). A Value-added Physical Properties Catalog for Low-redshift Galaxies from DESI Legacy Imaging Surveys DR10. The Astrophysical Journal Supplement Series, 284(2), 45.
- **Shirui Wei** et al. (2025). Photometric redshift estimation for emission-line galaxies in the DESI Legacy Imaging Surveys using CNN–MLP models. Publications of the Astronomical Society of Australia, 42, e092.
- **Shirui Wei** et al. (in preparation). In Here: Dwarf Galaxies in the OutThere Survey.
- **Shirui Wei** et al. (in preparation). A Scalable Pipeline for Multimodal Galaxy Property Inference: A Case Study with DESI LS DR10

Co-Author Papers:

- Wujun Shao, ..., **Shirui Wei** et al. (2025). Deep learning-based astronomical multimodal data fusion: A comprehensive review. Information Fusion, 104103.
  A comprehensive review of multimodal data and deep learning techniques in astronomy. Contributed to writing.
- Chao Tang, ..., **Shirui Wei** et al. (2025). A retrieval-augmented analysis framework for target galaxy search. Research in Astronomy and Astrophysics, 25(6), 065011.
  Developed a retrieval-augmented (RAG-like) framework for galaxy search and analysis. Contributed to experiments and writing.
- Changhua Li, ..., **Shirui Wei** et al. (2024). A photometric redshift catalogue of galaxies from the DESI Legacy Imaging Surveys DR10. The Astronomical Journal, 168(6), 233.
  Constructed a photometric redshift catalog for DESI LS DR10 based on EAZY and machine learning methods. Contributed to experiments.
- Shenglin Zhang, ..., **Shirui Wei** et al. (2024). No more data silos: Unified microservice failure diagnosis with temporal knowledge graphs. IEEE Transactions on Services Computing.
  Applied temporal knowledge graphs to integrate multimodal monitoring data for microservice failure diagnosis. Contributed to experiments and writing.

CONFERENCES

2025 Annual Meeting of the Chinese Astronomical Society | Xiamen, China Nov. 2025
Talk: Photometric Redshift Estimation of Emission-Line Galaxies Based on Multimodal Methods.

MPIA Galaxy Coffee | Heidelberg, Germany Sep. 2025
Talk: In Here: Dwarf Galaxies in the OutThere Survey.

Astroinformatics and China-VO 2025 | Haikou, China Nov. 2025
Poster & Flash Talk: Photometric Redshift Estimation of Emission-Line Galaxies Based on Multimodal Methods.

Astroinformatics and China-VO 2024 | Xinchang, Zhejiang Nov. 2024
Poster & Flash Talk: Photometric Redshift Catalogue of Galaxies from the DESI Legacy Imaging Surveys DR10.

COMPUTING

Astronomical Large Language Models Platform Contributor
A platform for the utilization of astronomical large language models, jointly provided by NAOC and Zhejiang Lab.

LAMOST ChatBot Developer
A Retrieval-Augmented Generation (RAG) based chatbot for LAMOST data release documents and value-added catalogs.

TEACHING

Astronomical Data Reduction | UCAS Mar. 2025 - May. 2025
Teaching Assistant with Prof. Yongheng Zhao. Assisted with course activities and software support, and led two observational internships at Xinglong Observatory, NAOC.

SKILLS

Computer skills

Languages: Python & C++

Techniques: Proficient in Astronomical Data Reduction Tools and Libraries (e.g. Astropy, Topcat), Deep Learning and LLM Frameworks (e.g. PyTorch, Langchain).

Languages

Chinese (Native), English, German (beginner)

AWARDS AND SCHOLARSHIPS

MPIA Summer Internship, Max Planck Institute for Astronomy	2025
Excellent Student, National Astronomical Observatories, Chinese Academy of Sciences	2025
Scholarship of Academic Excellence, Nankai University	2021-2022,2020-2021
Innovation Scholarship, Nankai University	2020-2021
Scholarship of Public Interests and All-Round Capability, Nankai University	2019-2020